

Section: Particles and Waves

Topic: Refraction of Light

A ray of light passes from air into a transparent block.

(a) State what happens to the speed and wavelength of the light as it enters the block.

✔ As light enters a denser medium (e.g. from air to glass):

- Speed decreases
- Wavelength decreases

The frequency stays the same, but since $v = f\lambda$, a decrease in speed results in a shorter wavelength.

(b) State what happens to the frequency of the light as it enters the block.

✔ The frequency remains constant.

The source determines the frequency, and it does **not change** when light passes into a different medium.

(c) The angle of incidence is 45° .

The angle of refraction is 29° .
Calculate the refractive index of the material.

Use Snell's Law:

$$n = \frac{\sin i}{\sin r} = \frac{\sin(45^\circ)}{\sin(29^\circ)} = \frac{0.7071}{0.4848} = 1.46$$

🧠 Revision Tips

- In denser media: speed ↓, wavelength ↓, frequency = constant
- Use Snell's Law:

$$n = \frac{\sin i}{\sin r}$$

- A higher refractive index means light slows down more
- Always use degrees in radians mode only if explicitly stated